LARCH

AUTOMOTIVE MANUFACTURING SOFTWARE

Production Planning & Control

Enterprise-Grade Software Capabilities for Modern Automotive Production

MRP | APS | FCS | MES | KPI



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What's Inside

01

Production Planning & MRP

Demand-driven, time-phased scheduling

02

Advanced Planning & Optimization

Multi-objective constraint-based APS

03

Work Order & FCS

Shop floor dispatching & finite capacity

04

MES Integration & Analytics

Real-time execution & KPI dashboards

05

Quality & Supplier Sync

Compliance, traceability & collaboration

06

Integration & Connectivity

ERP, PLM, IoT, APIs & EDI

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01

Production Planning & MRP

Demand-driven scheduling across multiple planning horizons

01

1 Demand-Driven Scheduling

Forecast vs actual demand comparison with sales order prioritization rules

2 Time-Phased Planning

Daily, weekly & monthly plans with short, mid & long horizon views

3 Capacity Checks

Finite vs infinite scheduling with shop floor capacity balancing

4 Multi-Site Synchronization

Central & distributed scheduling coordination across facilities

5 Automated MRP

Net requirements calculation, multi-level BOM explosion & pegging analysis

6 Exception Messaging

Priority alerts for shortages, delays & corrective action suggestions

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02

Advanced Planning & Optimization

Multi-objective APS with real-time rescheduling capability

02

1 Multi-Objective Optimization

Minimize makespan, changeover costs & inventory simultaneously

2 Constraint Modeling

Machine setups, changeover times, tooling & operator skill constraints

3 What-If Simulations

Model downtime impacts, new product introductions & alternate routes

4 Real-Time Rescheduling

Automatic schedule adjustments triggered during production disruptions

5 Resource Optimization

Simultaneous material and resource planning for maximum efficiency

6 Scenario Planning

Alternate supplier & routing scenario analysis for strategic risk control

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03

Work Order & Finite Capacity

End-to-end job management with true capacity visibility

03

1 Work Order Management

Auto creation from MPS/MRP with flexible sequencing rules

2 Routing & Operations

Standard routing templates with operation splitting & merging support

3 Shop Floor Dispatching

Digital work tickets with Kanban & pull-based scheduling support

4 True Capacity Visibility

Machine, tooling & labor calendars for accurate capacity planning

5 Load Leveling

Smooth out utilization peaks to optimize throughput across lines

6 Bottleneck Analysis

Identify constraints with Gantt drag-and-drop schedule editing tools

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04

MES Integration & Analytics

Real-time shop floor execution powering performance insight

04

1 Real-Time Data Capture

Barcode operations, PLC/IoT machine interfaces & live production status

2 Production Dashboard

Completed vs scheduled views, OEE metrics & instant exception alerts

3 Labor Tracking

Employee time & attendance per job with automated skill matching

4 Performance Indicators

On-time delivery, throughput, cycle time & WIP level monitoring

5 Quality Outcomes

Scrap & rework rates per operation with defect classification trending

6 Predictive Analytics

Forecast delays using historical patterns before disruptions occur

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05

Quality & Supplier Sync

Integrated compliance, traceability & collaborative planning

05

1 In-Process Checks

Inspection plans tied to operations with SPC dashboards & control charts

2 Non-Conformance & CAPA

Automated alerts & structured corrective action workflow management

3 Traceability Linkage

Component lots & serials linked to finished vehicles for recall readiness

4 Regulatory Audit Support

Digital logs with full version control for compliance & certification

5 Supplier Visibility

Shared planning views with capacity & lead-time transparency for vendors

6 Vendor Performance

On-time delivery & quality scoring with inbound ASN dock scheduling

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Integration & Digital Connectivity

ERP Linkage

Seamless connection to finance, costing & order management modules

PLM / BOM Sync

Engineering change control with automated BOM update propagation

APIs & EDI

Open API & EDI standards for supplier and customer system connectivity

IoT & Sensors

Machine health monitoring & predictive maintenance via sensor integration

CAD / BOM Interchange

Automated BOM updates from engineering design changes

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Built for Automotive Complexity

AUTOMOTIVE CHALLENGE	→	SOFTWARE CAPABILITY
Complex Product Variants	→	Multi-level BOM explosion + MPS planning
Tight Delivery Windows	→	Real-time scheduling & auto rescheduling
Supplier Variability	→	Collaborative planning & lead time management
Quality & Compliance	→	Integrated in-process checks & audit trails
High Cost of Downtime	→	Finite capacity + APS optimization engines